

Similarly we can show that when both $\neg p, \neg q$ are False, both p, q are True.

$\therefore p \leftrightarrow q$ and $\neg p \leftrightarrow \neg q$ are logically equivalent $\square$

34) Show that $(p \vee q) \wedge (\neg p \vee \neg r) \rightarrow (q \vee r)$ is a tautology.

Soln: Let the given compound proposition not be a tautology. By the defn of not a tautology, there exists an assignment of truth values of p, q and r such that $(p \vee q) \wedge (\neg p \vee \neg r) \rightarrow (q \vee r)$ is False. By the defn. of conditional, $(p \vee q) \wedge (\neg p \vee \neg r)$ is True and $(q \vee r)$ is False. By the defn. of OR, both q, r is False by the defn. of AND, $(p \vee q)$ is True and $(\neg p \vee \neg r)$ is True. Since both q, r is False, by the defn. of OR, p is True and $\neg p$ is True. But this is a contradiction since by the defn. of negation if p is True, $\neg p$ is False.

$\therefore$ It is a tautology $\square$

35) Show that $(p \rightarrow q) \rightarrow \neg r$ and $p \rightarrow (q \rightarrow r)$ are not logically equivalent.

Soln: Let, $(p \rightarrow q) \rightarrow \neg r$ be ~~True~~ False. By defn. of conditional, $(p \rightarrow q)$ is True and r is False. By defn. of conditional, 3 cases arise:
① p is True, q is True ② p is False, q is False ③ p is False, q is True.

Case 1: p is True, q is True, r is False. By defn. of conditional, $q \rightarrow r$ is False. By defn. of conditional, $p \rightarrow (q \rightarrow r)$ is False.

Case 2: p is False, q is False, r is True. By defn. of conditional, $p \rightarrow (q \rightarrow r)$ is True. But $(p \rightarrow q) \rightarrow \neg r$ is False.

$\therefore (p \rightarrow q) \rightarrow \neg r$ and $p \rightarrow (q \rightarrow r)$ are not logically equivalent $\square$

36) Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent.

Soln: Let $(p \wedge q) \rightarrow r$ be False. By defn. of conditional, $p \wedge q$ is True and r is False. By defn. of AND, p is True, q is True. By defn. of conditional, $(p \rightarrow r)$ is False, $(q \rightarrow r)$ is False. By defn. of AND, $(p \rightarrow r) \wedge (q \rightarrow r)$ is False.

Let $(p \wedge q) \rightarrow r$ be True. The following cases arise: ① r is True. Here, the truth values of $p \wedge q$ doesn't matter. By defn. of conditional, $(p \rightarrow r)$ and $(q \rightarrow r)$ are both respectively True. By defn. of